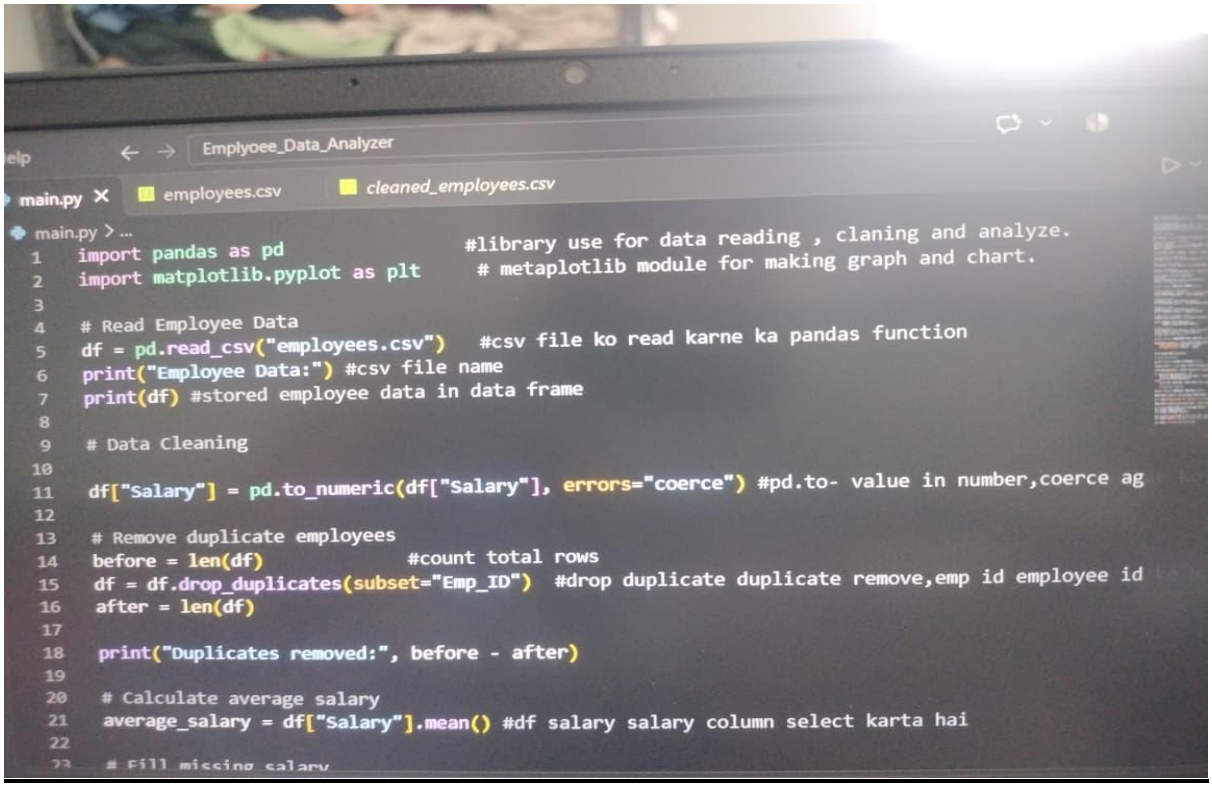
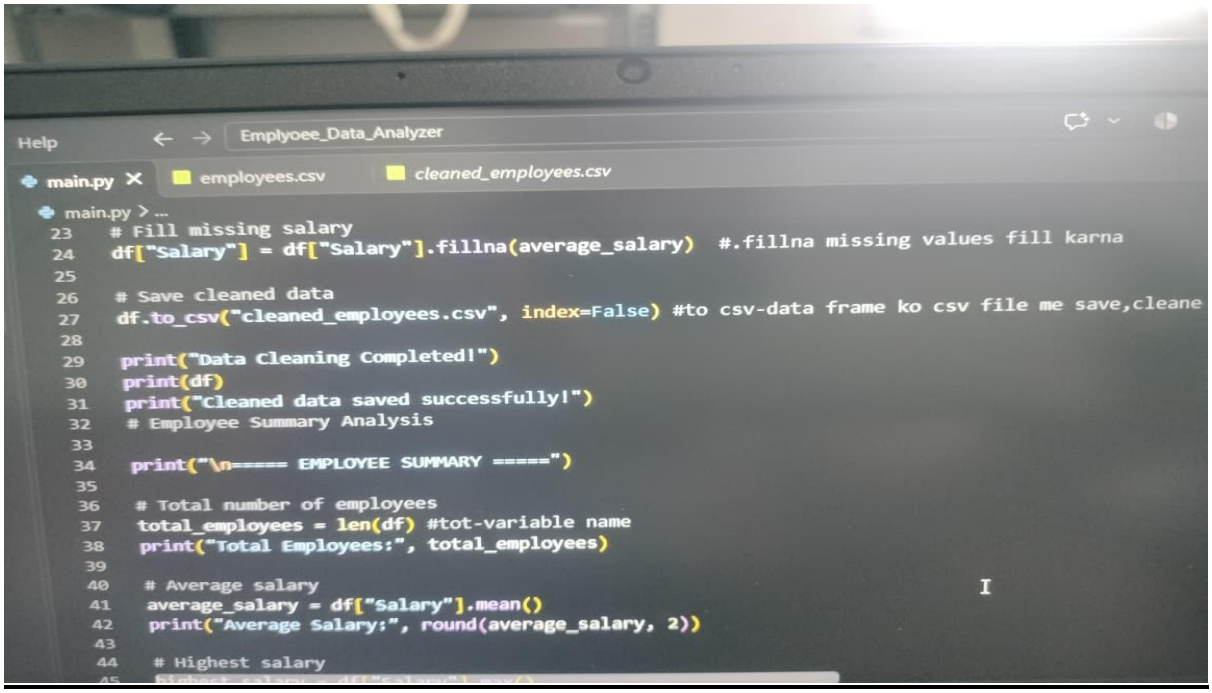
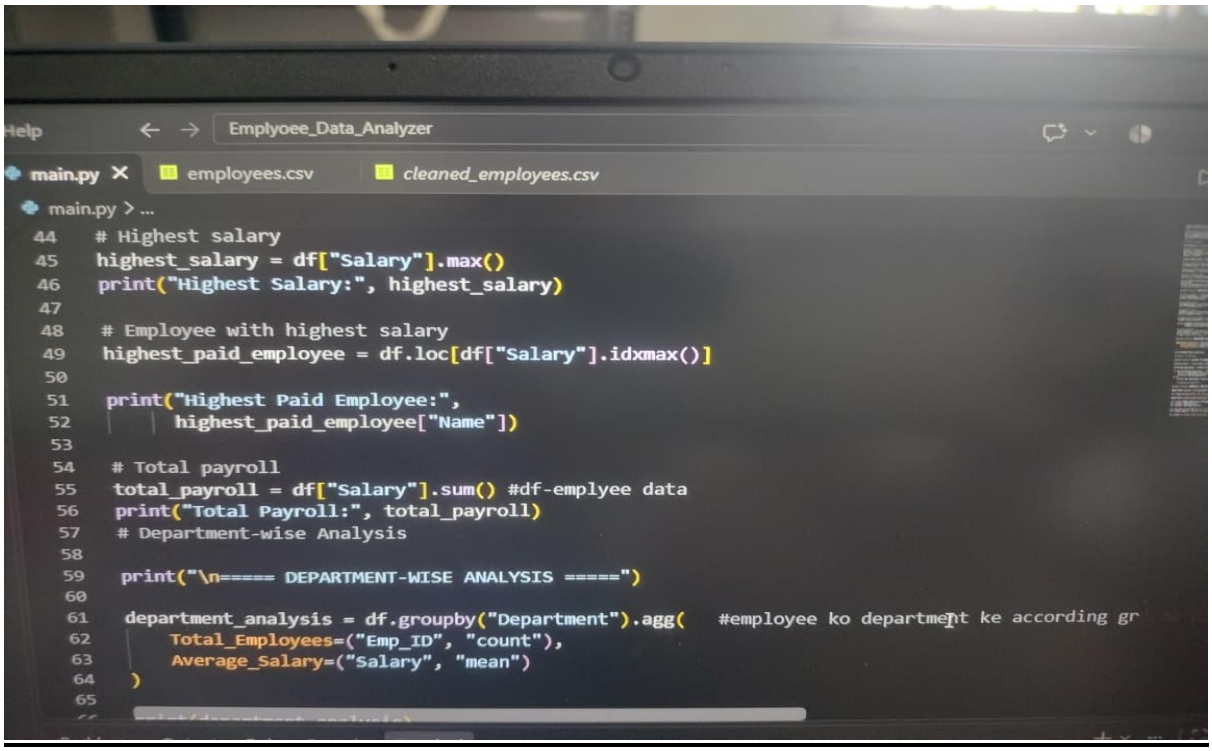
# EMPLOYEE DATA ANALYZER.DOCX

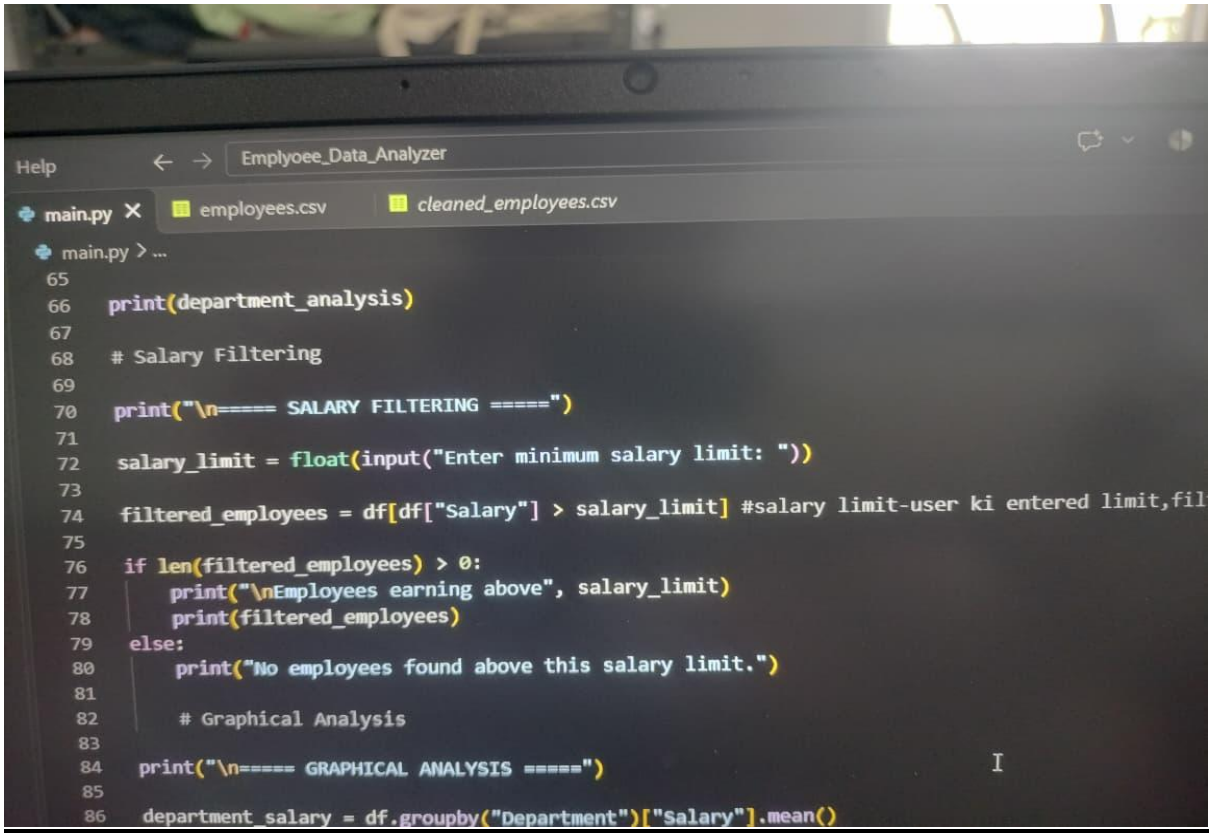
# EMPLOYEE DATA ANALYZER

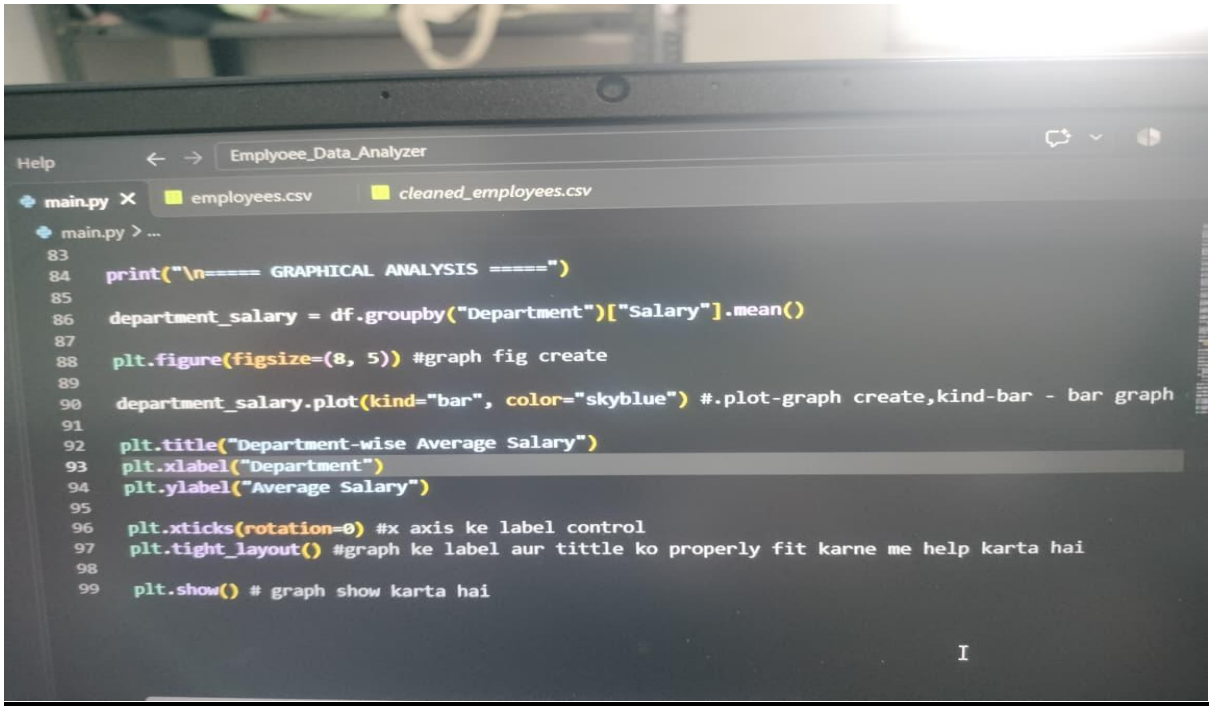
## CODE:



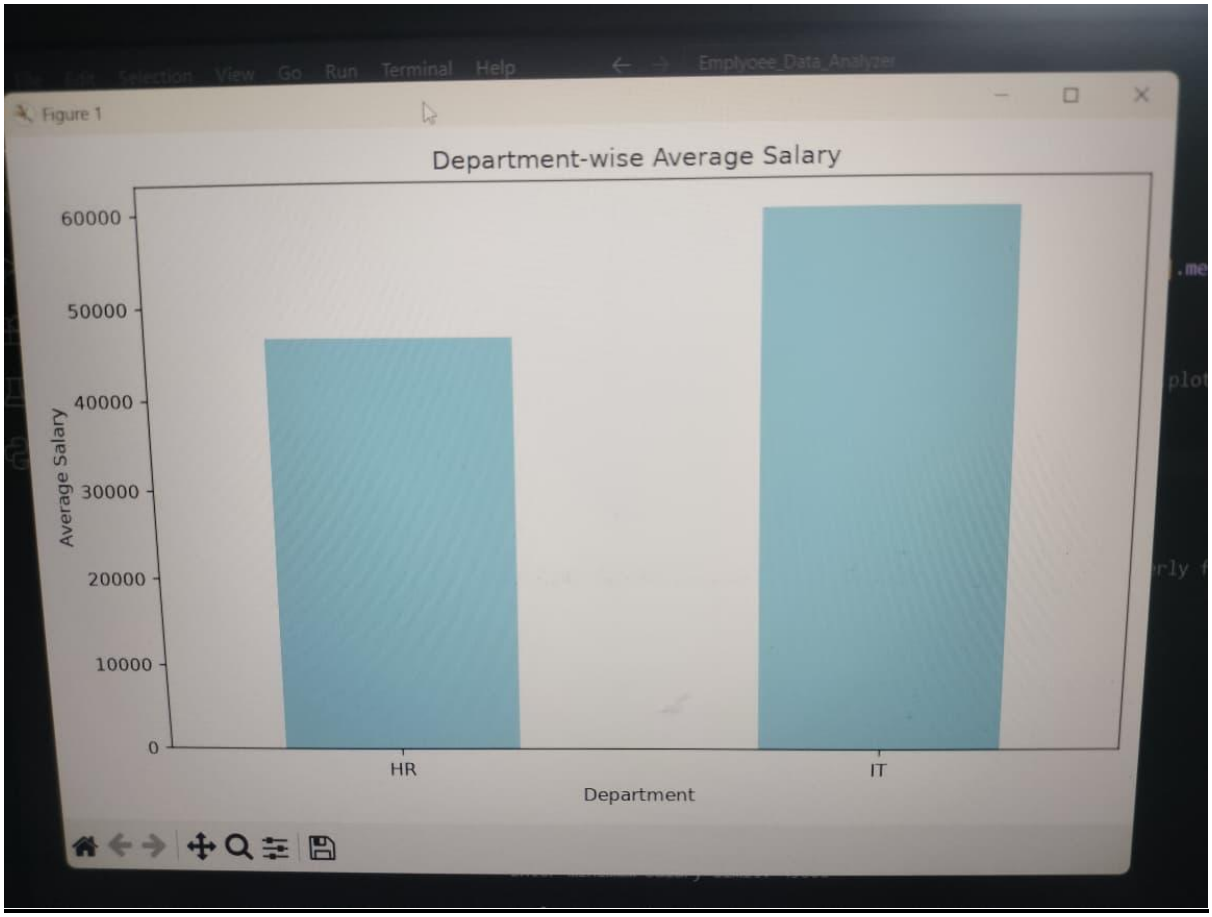

Help
Emplyoee_Data_Analyzer
main.py
employees.csv
cleaned_employees.csv
main.py > ...
# Fill missing salary
df["Salary"] = df["Salary"].fillna(average_salary) #.fillna missing values fill karna
# Save cleaned data
df.to_csv("cleaned_employees.csv", index=False) #to csv-data frame ko csv file me save,cleane
print("Data Cleaning Completed!")
print(df)
print("Cleaned data saved successfully!")
# Employee Summary Analysis
print("\n===== EMPLOYEE SUMMARY =====")
# Total number of employees
total_employees = len(df) #tot-variable name
print("Total Employees:", total_employees)
# Average salary
average_salary = df["Salary"].mean()
print("Average Salary:", round(average_salary, 2))
# Highest salary


Help
Emplyoee_Data_Analyzer
main.py
employees.csv
cleaned_employees.csv
main.py > ...
# Highest salary
highest_salary = df["Salary"].max()
print("Highest Salary:", highest_salary)
# Employee with highest salary
highest_paid_employee = df.loc[df["Salary"].idxmax()]
print("Highest Paid Employee:",
highest_paid_employee["Name"])
# Total payroll
total_payroll = df["Salary"].sum() #df-emplyee data
print("Total Payroll:", total_payroll)
# Department-wise Analysis
print("\n===== DEPARTMENT-WISE ANALYSIS =====")
department_analysis = df.groupby("Department").agg( #employee ko department ke according gr
Total_Employees=("Emp_ID", "count"),
Average_Salary=("Salary", "mean")
)

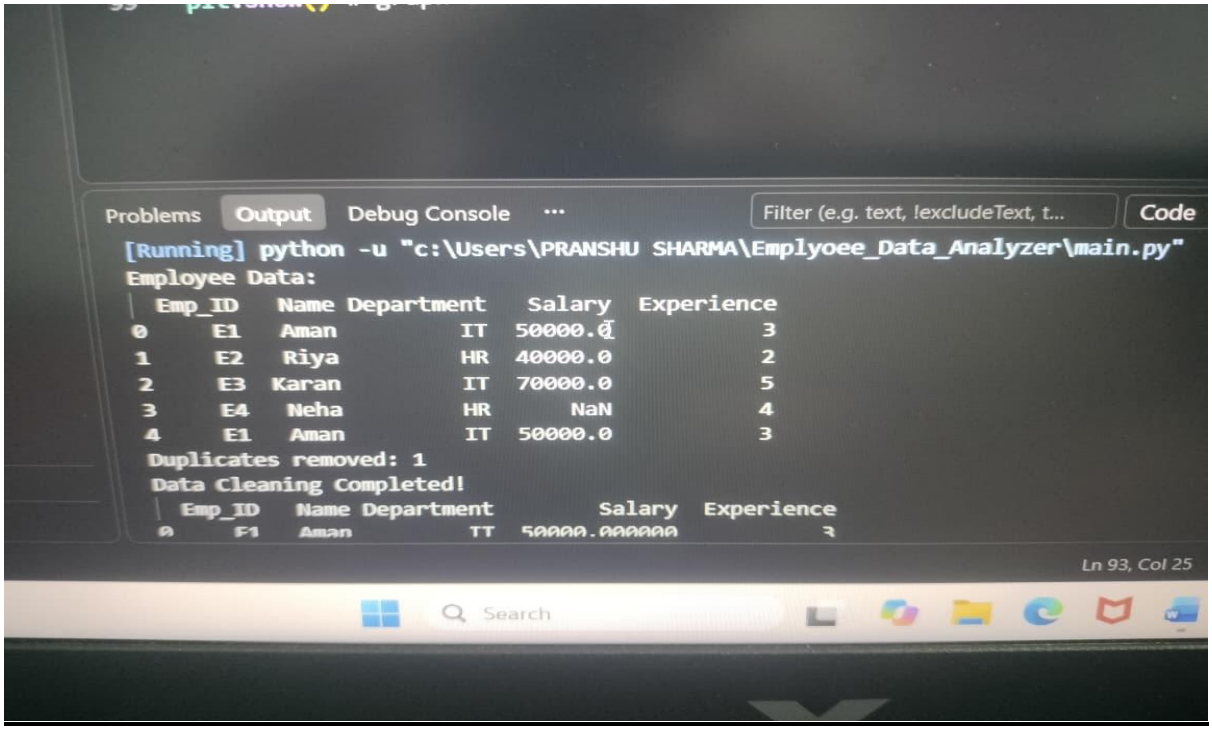

Help
Emplyoee_Data_Analyzer
main.py
employees.csv
cleaned_employees.csv
main.py > ...
65
66 print(department_analysis)
67
68 # Salary Filtering
69
70 print("\n===== SALARY FILTERING =====")
71
72 salary_limit = float(input("Enter minimum salary limit: "))
73
74 filtered_employees = df[df["Salary"] > salary_limit] #salary limit-user ki entered limit,fil
75
76 if len(filtered_employees) > 0:
77 print("\nEmployees earning above", salary_limit)
78 print(filtered_employees)
79 else:
80 print("No employees found above this salary limit.")
81
82 # Graphical Analysis
83
84 print("\n===== GRAPHICAL ANALYSIS =====")
85
86 department_salary = df.groupby("Department")["Salary"].mean()

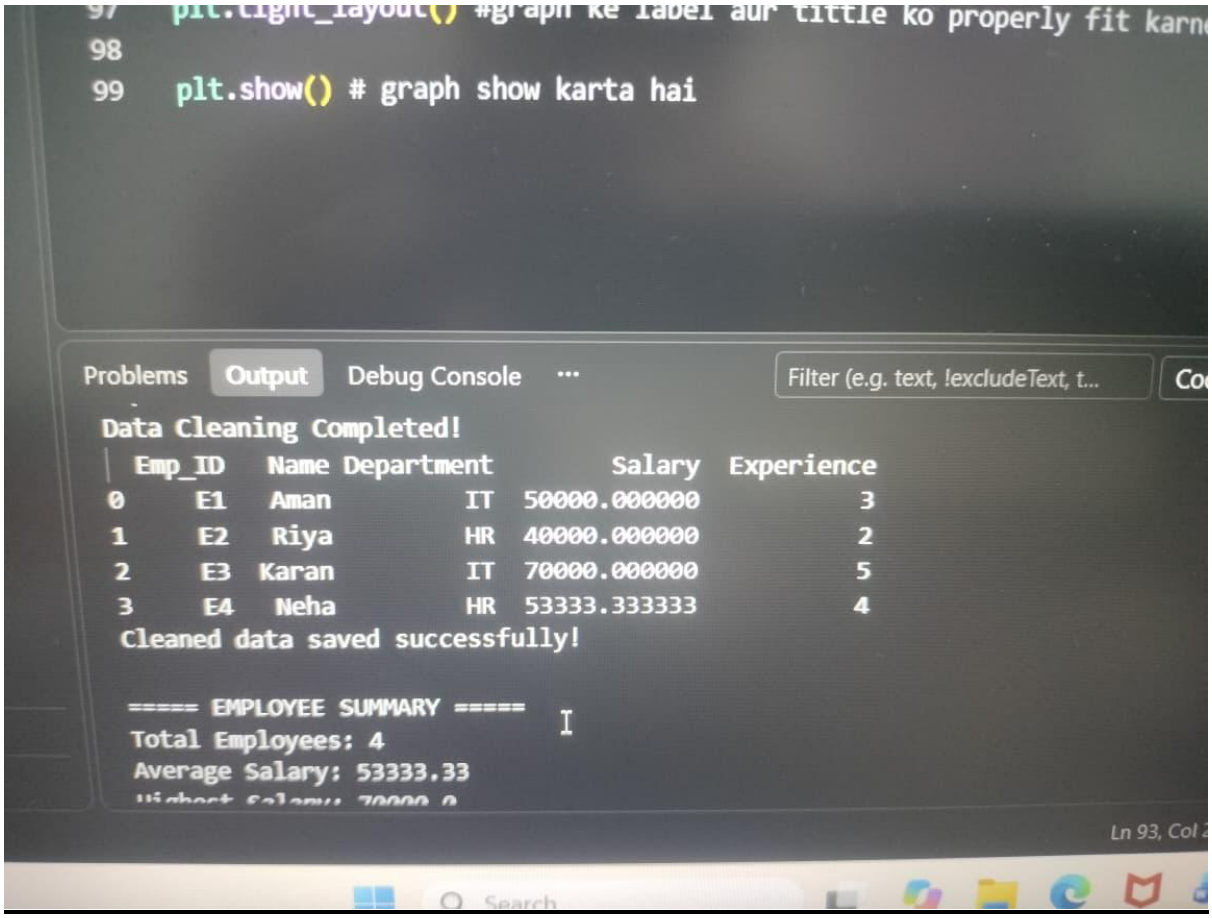

Help
Emplyoee_Data_Analyzer
main.py
employees.csv
cleaned_employees.csv
main.py > ...
83
84 print("\n===== GRAPHICAL ANALYSIS =====")
85
86 department_salary = df.groupby("Department")["Salary"].mean()
87
88 plt.figure(figsize=(8, 5)) #graph fig create
89
90 department_salary.plot(kind="bar", color="skyblue") #.plot-graph create,kind-bar - bar graph
91
92 plt.title("Department-wise Average Salary")
93 plt.xlabel("Department")
94 plt.ylabel("Average Salary")
95
96 plt.xticks(rotation=0) #x axis ke label control
97 plt.tight_layout() #graph ke label aur tittle ko properly fit karne me help karta hai
98
99 plt.show() # graph show karta hai

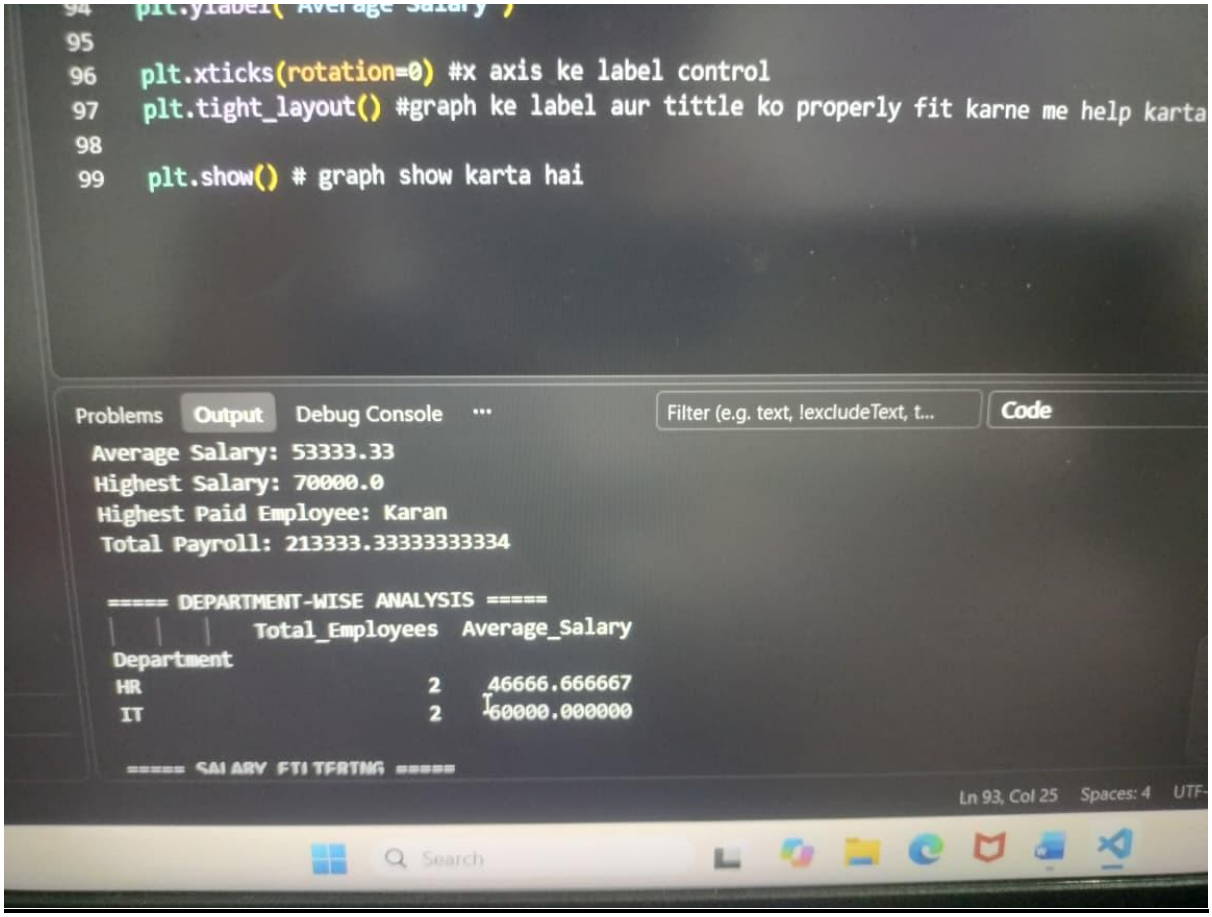
## GRAPH:

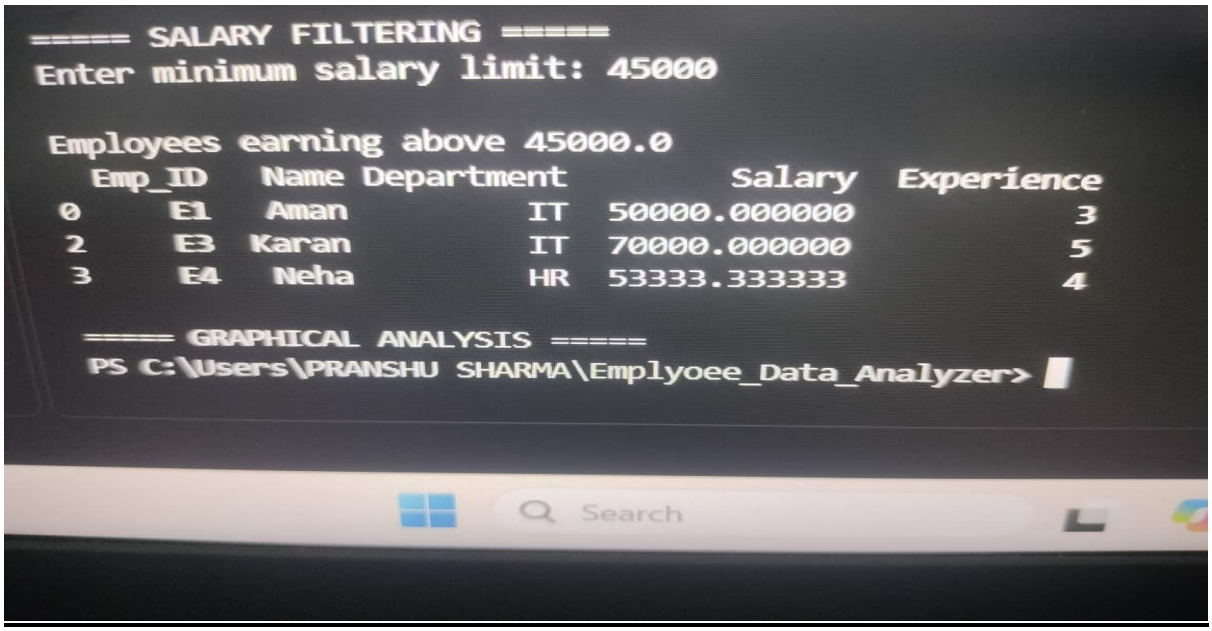

Figure 1
Department-wise Average Salary
60000
50000
40000
30000
20000
10000
0
Average Salary
HR
IT
Department

## OUTPUT:

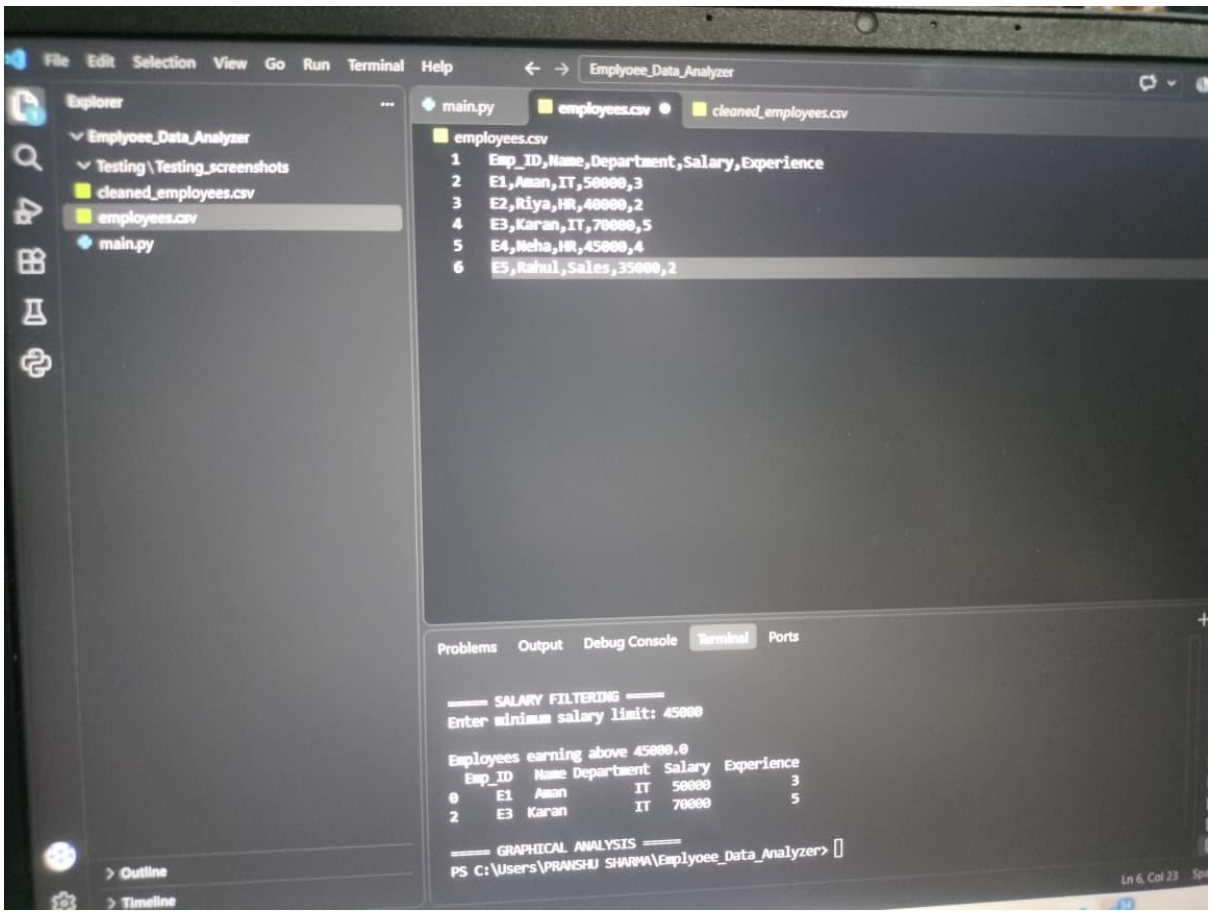

Problems
Output
Debug Console
Filter (e.g. text, !excludeText, t...
Code
[Running] python -u "c:\Users\PRANSHU SHARMA\Emplyoee_Data_Analyzer\main.py"
Employee Data:
Emp_ID Name Department Salary Experience
0 E1 Aman IT 50000.0 3
1 E2 Riya HR 40000.0 2
2 E3 Karan IT 70000.0 5
3 E4 Neha HR NaN 4
4 E1 Aman IT 50000.0 3
Duplicates removed: 1
Data Cleaning Completed!
Emp_ID Name Department Salary Experience
Ln 93, Col 25
Search

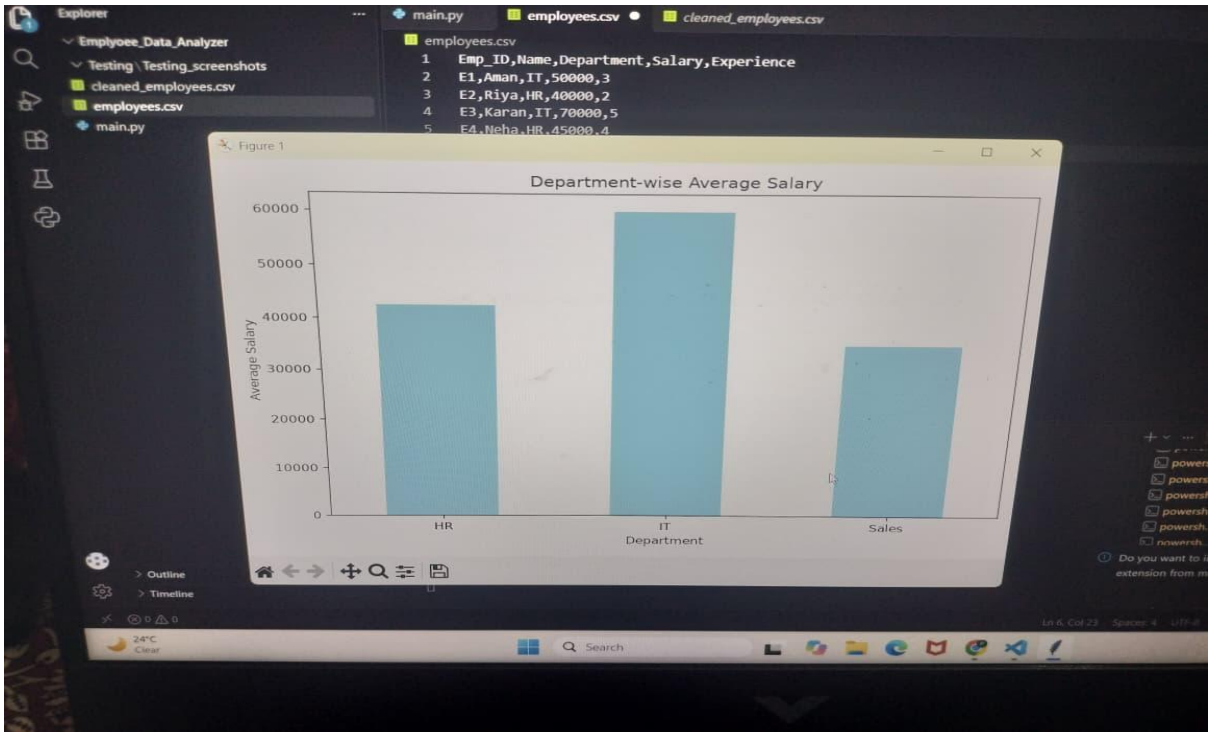

plt.tight_layout() #graph ke label aur title ko properly fit karne
plt.show() # graph show karta hai
Problems
Output
Debug Console
Filter (e.g. text, !excludeText, t...
Data Cleaning Completed!
Emp_ID Name Department Salary Experience
0 E1 Aman IT 50000.000000 3
1 E2 Riya HR 40000.000000 2
2 E3 Karan IT 70000.000000 5
3 E4 Neha HR 53333.333333 4
Cleaned data saved successfully!
===== EMPLOYEE SUMMARY =====
Total Employees: 4
Average Salary: 53333.33
Ln 93, Col
Search


plt.xticks(rotation=0) #x axis ke label control
plt.tight_layout() #graph ke label aur tittle ko properly fit karne me help karta
plt.show() # graph show karta hai
Problems
Output
Debug Console
Filter (e.g. text, !excludeText, t...
Code
Average Salary: 53333.33
Highest Salary: 70000.0
Highest Paid Employee: Karan
Total Payroll: 213333.33333333334
===== DEPARTMENT-WISE ANALYSIS =====
Total_Employees Average_Salary
Department
HR 2 46666.666667
IT 2 60000.000000
Ln 93, Col 25
Spaces: 4
Search

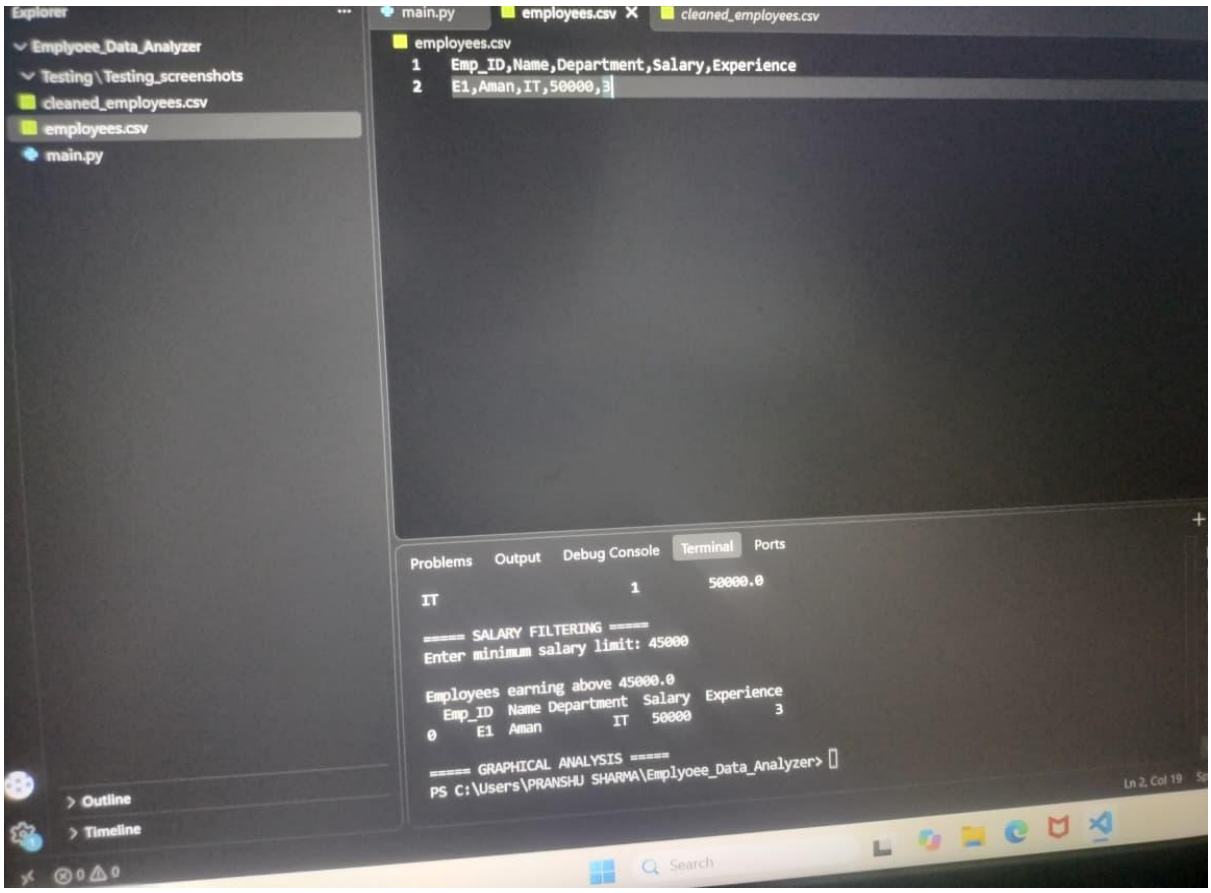

===== SALARY FILTERING =====
Enter minimum salary limit: 45000
Employees earning above 45000.0
Emp_ID Name Department Salary Experience
0 E1 Aman IT 50000.000000 3
2 E3 Karan IT 70000.000000 5
3 E4 Neha HR 53333.333333 4
===== GRAPHICAL ANALYSIS =====
PS C:\Users\PRANSHU SHARMA\Emplyoee_Data_Analyzer>
Search

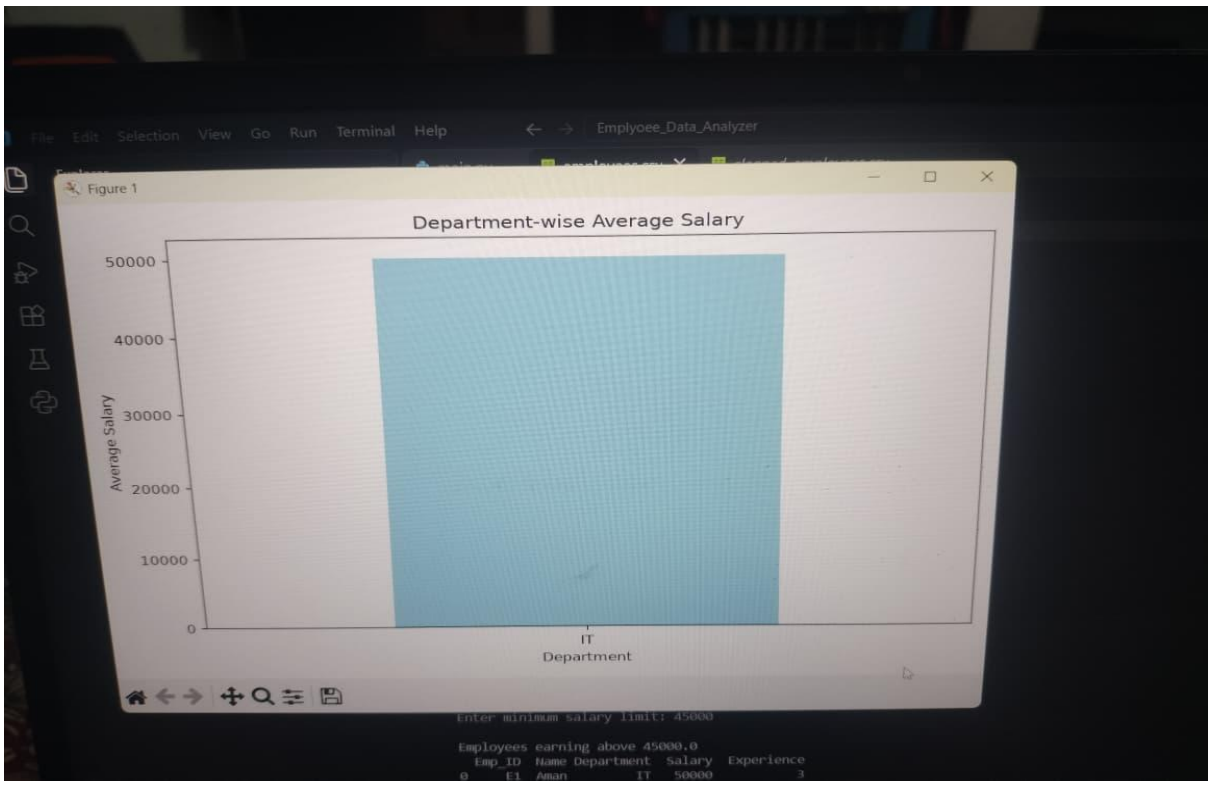
# Test case1:-Original employee data testing



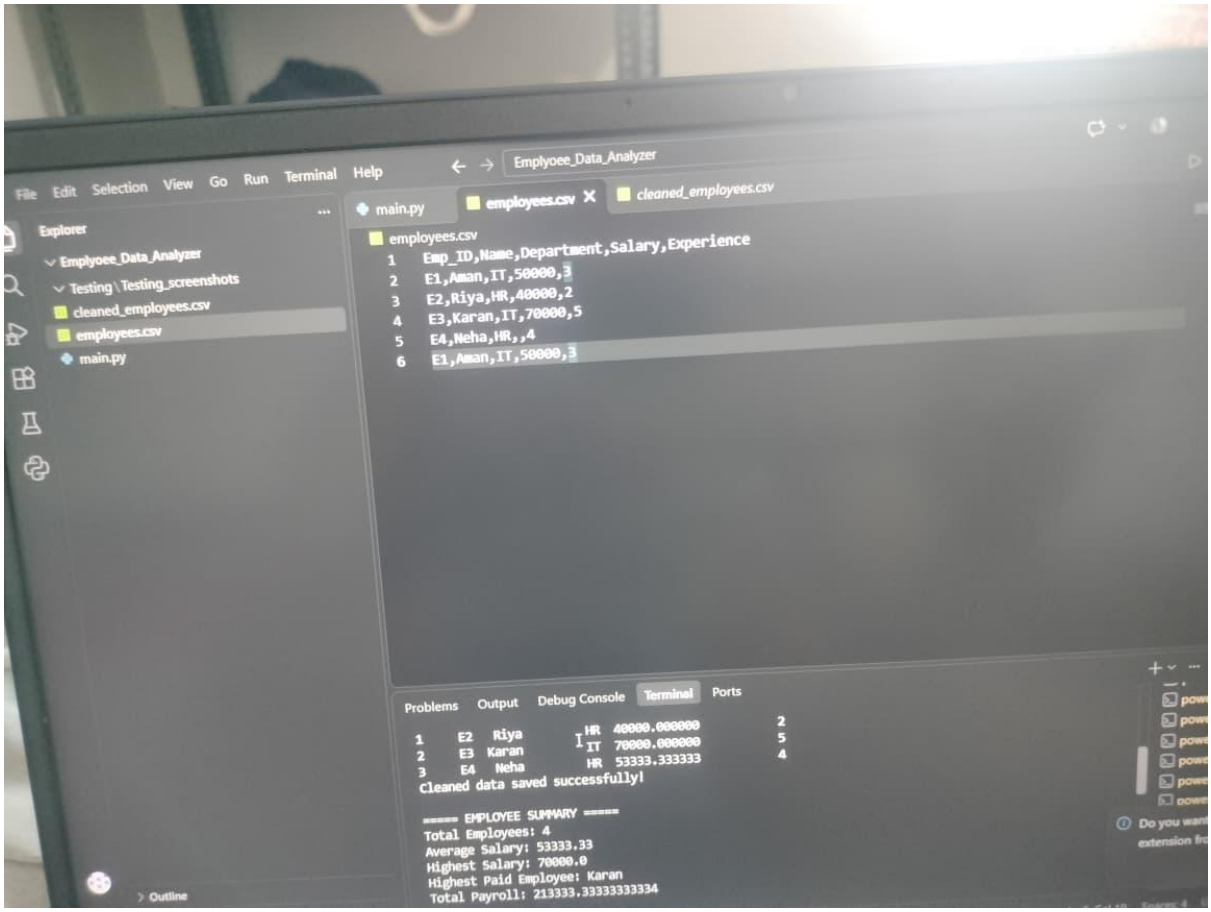



# Test case2: Single Employee Data Testing



# Test case3: Duplicate and missing salary



# Test case4: Department wise analysis testing

